

ALLWYN NOBLE

Software Engineering | AI/ML | Full-Stack Development

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EDUCATION

Karunya Institute of Technology and Sciences, Coimbatore, Tamil Nadu

June 2023 – Present

B.Tech in Computer Science Engineering (Artificial Intelligence)

Expected Graduation: 2027

Suguna RIP V School, Coimbatore, Tamil Nadu

June 2021 – May 2023

Higher Secondary Certificate (HSC)

TECHNICAL SKILLS

Languages:	Python, Java, Kotlin, JavaScript, SQL, PHP
AI/ML:	Machine Learning, Federated Learning, PyTorch, Scikit-learn, NLP, SMOTE
Frameworks:	FastAPI, React, Next.js, Android, Jetpack Compose, REST APIs
Databases:	PostgreSQL, SQLite, MySQL, Supabase
Tools:	Git, GitHub, Android Studio, Streamlit, Vercel

PROJECTS

Energy-Aware Federated Learning for Healthcare *Python • PyTorch • Flower • FastAPI • Kotlin • Android • Jetpack Compose • SQLite*

- **Engineered** a privacy-preserving healthcare screening framework using **Flower Federated Learning (FedAvg/FedAsync)** and **PyTorch** for low-connectivity environments, ensuring patient data remains on edge devices.
- **Built** a native **Android healthcare client** in **Kotlin** and **Jetpack Compose**, implementing offline-first patient screening with local **Room storage** and **WorkManager** background sync upon connectivity recovery.
- **Developed** an energy-aware **Deep Q-Network (DQN)** scheduler on a **FastAPI** & **SQLite** backend to dynamically optimize model aggregation using real-time device battery, network latency, and bandwidth telemetry.
- **Integrated** a **Healthcare Edge Control Center** for real-time monitoring of federated learning status, edge node telemetry, patient screening risk classifications, and critical emergency alert handling.

Full-Stack Dealership & Service Management Platform

Next.js • React • Supabase • PostgreSQL • REST APIs • Git • Vercel

- **Engineered** a full-stack dealership platform (for Kovai Motobikes) using **Next.js**, **React**, and **PostgreSQL** via **Supabase**, deployed live on **Vercel**.
- **Implemented** dynamic vehicle catalogue workflows featuring multi-criteria search, engine-capacity filtering, price sorting, service booking, test-ride scheduling, and direct **WhatsApp** sales integration.
- **Developed** an authenticated administrative dashboard using **Supabase Authentication** to securely manage vehicle inventory, track service bookings, and handle customer enquiries.
- **Built** database-backed **REST API** workflows and responsive mobile-friendly UI, integrating **Google Maps API** for interactive showroom location details.

CERTIFICATIONS

- **Machine Learning Workshop** – IIT Madras
- **Cyber Threat** – Coursera
- **Generative AI** – LinkedIn Learning
- **Intermediate Machine Learning** – Kaggle

RELEVANT COURSEWORK

- Machine Learning
- Artificial Intelligence
- Database Management Systems
- Computer Networks
- Natural Language Processing
- Software Engineering